

FORM TP 2010236



TEST CODE **02238020**

MAY/JUNE 2010

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

PHYSICS

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of **SIX** questions.
2. Section A consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in the spaces provided in this question paper.
3. Section B consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in the spaces provided in this question paper.
4. All working **MUST** be **CLEARLY** shown.
5. The use of non-programmable calculators is permitted, but candidates should note that the use of an inappropriate number of figures in answers will be penalised.

LIST OF PHYSICAL CONSTANTS

Speed of light in free space	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_0	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_0	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$ $\frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \text{ m f}^{-1}$
Elementary charge	e	=	$1.60 \times 10^{-19} \text{ C}$
Planck's constant	h	=	$6.63 \times 10^{-34} \text{ J s}$
Unified atomic mass constant (Energy equivalence: $1u = 93 \text{ Mev}$)	u	=	$1.66 \times 10^{-27} \text{ kg}$
Rest mass of electron	m_e	=	$9.11 \times 10^{-31} \text{ kg}$
Rest mass of proton	m_p	=	$1.67 \times 10^{-27} \text{ kg}$
Acceleration due to gravity	g	=	9.81 m s^{-2}
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$

SECTION A

Answer ALL questions.

You MUST write your answers in the spaces provided in this answer booklet.

1. (a) (i) Define the term 'capacitance'.

[1 mark]

- (ii) A capacitor with a capacitance of $2100\ \mu\text{F}$ is charged until the potential difference between its plates is $6.0\ \text{V}$. Determine

- a) the charge on one of the plates

[2 marks]

- b) the energy stored by the capacitor.

[2 marks]

- (b) A student set up the circuit shown in Figure 1 and after closing the switch, S, observed how the voltmeter reading varied with time. A graph of the data obtained is shown in Figure 2. Noting that the voltmeter reading decreased as time elapsed the student proposed the hypothesis that the voltmeter reading V is inversely proportional to the elapsed time t .

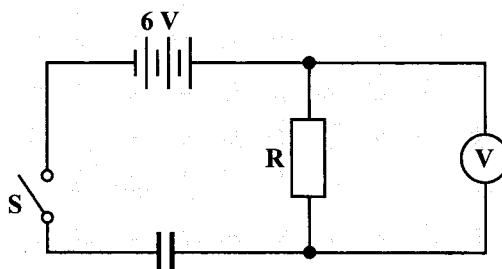


Figure 1

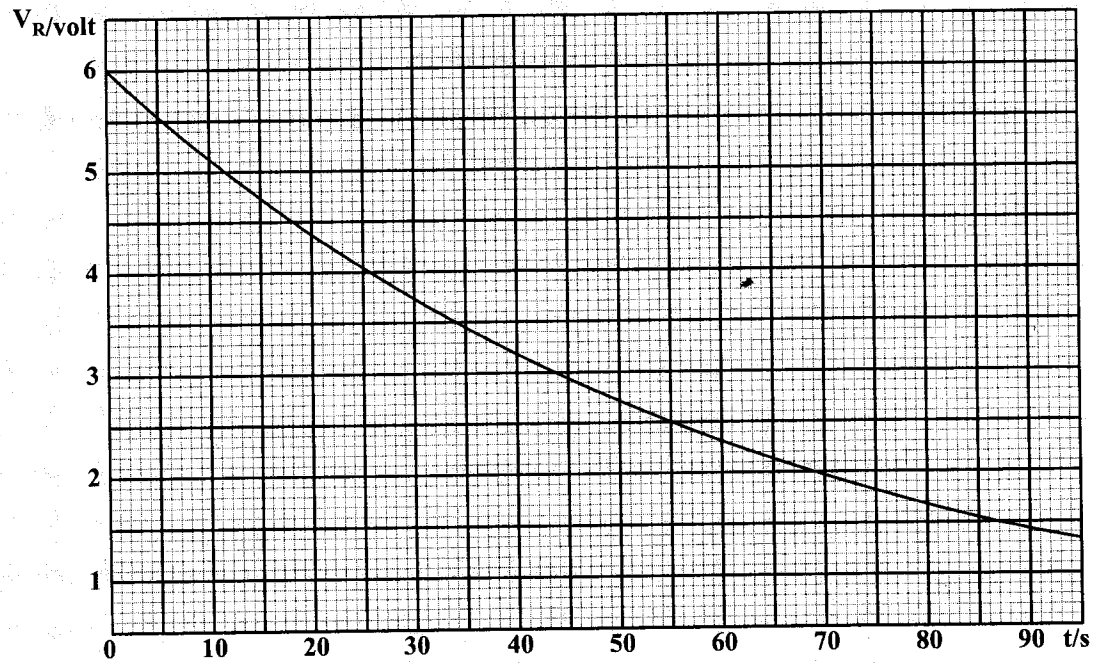


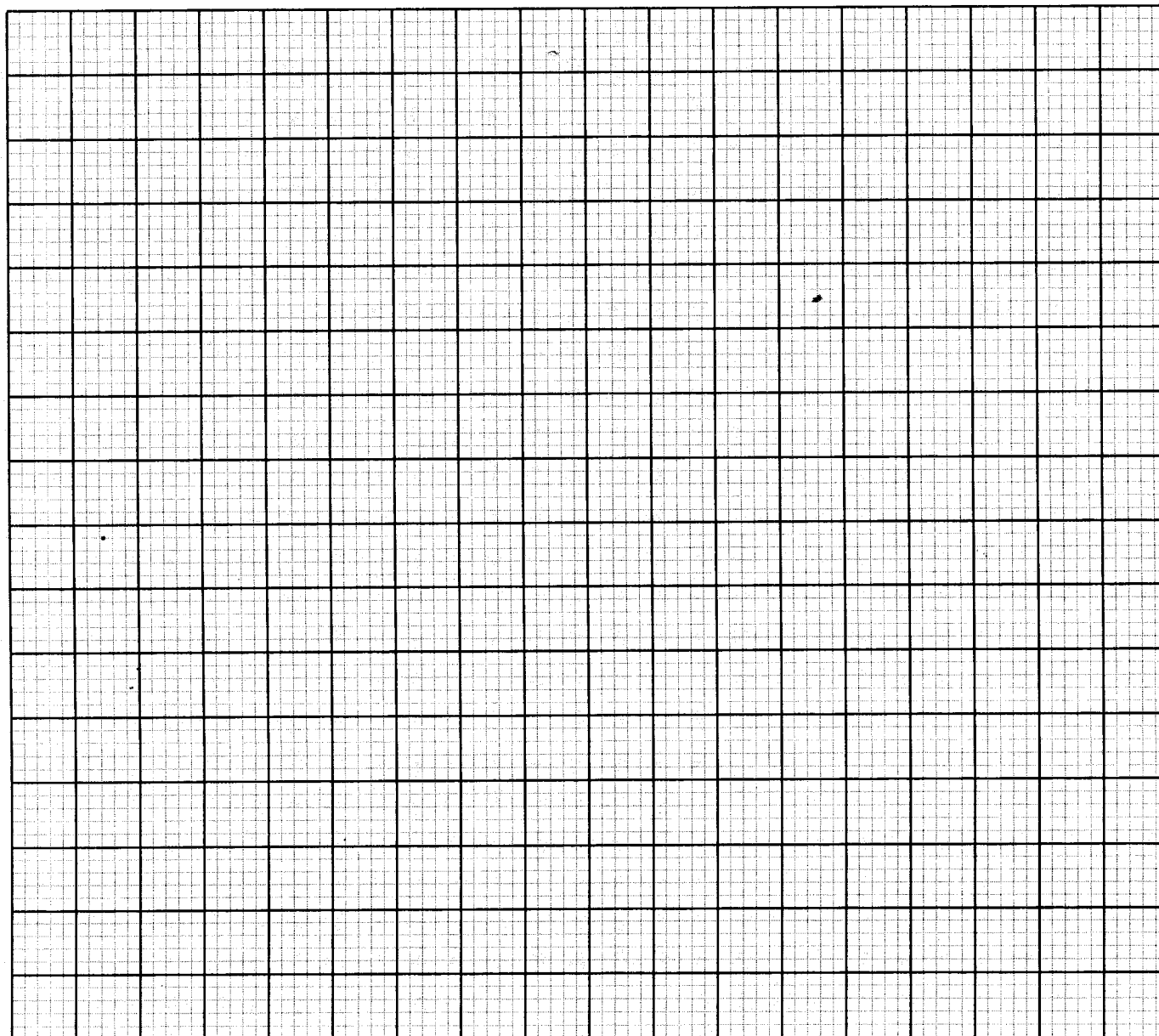
Figure 2. Graph of V_R vs Time

- (i) Test his hypothesis by taking data from the graph in Figure 2 to complete Table 1 and then plotting $1/V$ against t on the grid provided on page 5.

TABLE 1

t/s	V/V	$1/V$
15		
30		
45		
60		
75		
90		

[8 marks]



- (ii) State, with reasons, the conclusion which may be drawn from the data.

[2 marks]

Total 15 marks

2. (a) In relation to a semi-conductor

(i) explain what is meant by

a) P-type material

[1 mark]

b) N-type material

[1 mark]

c) the depletion region.

[1 mark]

(ii) Draw a diagram of a junction transistor and draw the transistor symbol.

[2 marks]

- (b) The current, I , through the p-n junction of a diode is related to the voltage across the junction, V , by the expression

$$I = I_0 \exp \left[\frac{e V}{n k T} \right]$$

where I_0 is the saturation current; n is the junction ideality factor; k is the Boltzman's constant and T is the temperature in kelvin. Some d.c. measurements made on a silicon diode at 300 K, yielded the following results:

TABLE 2

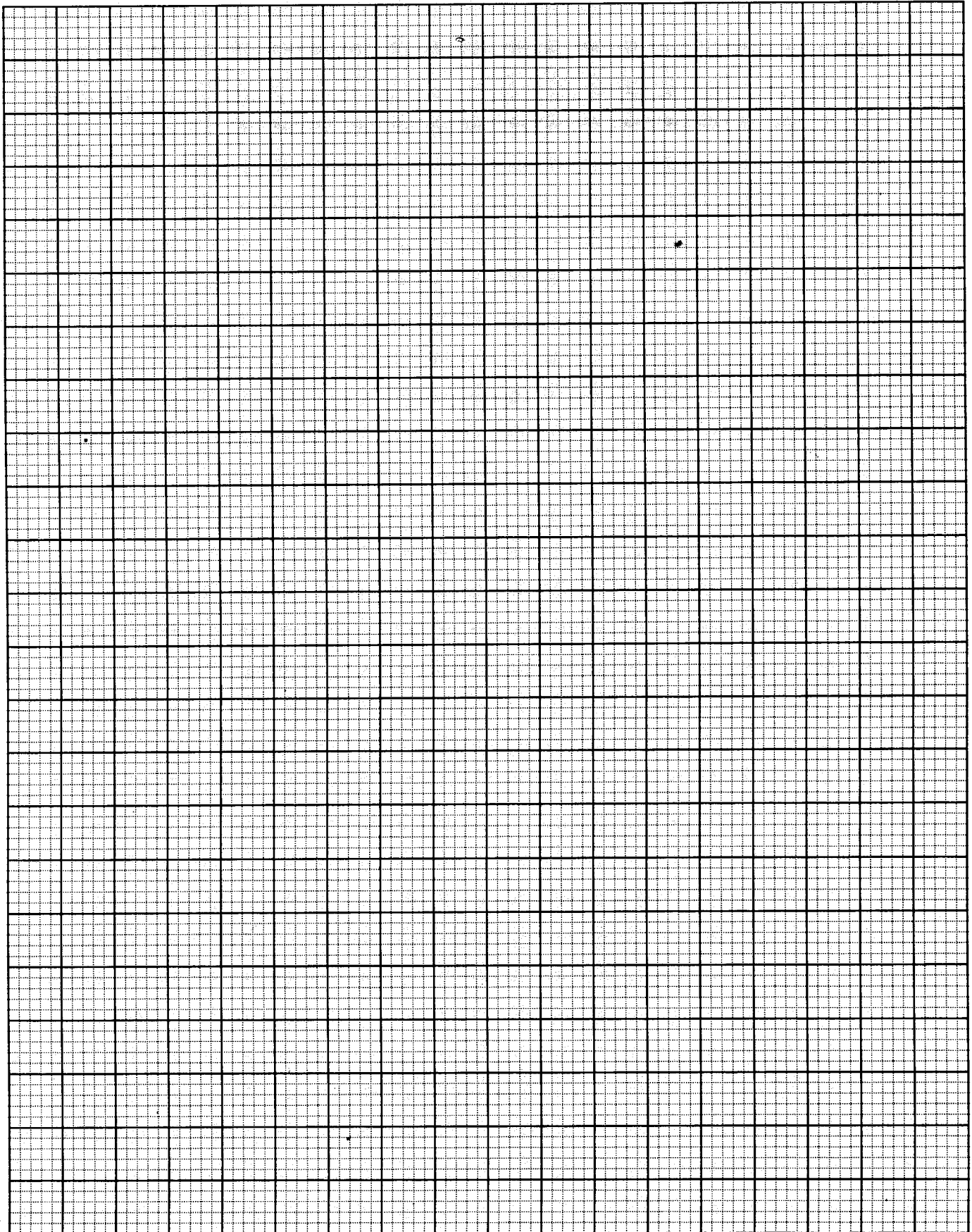
V/Volt	I/ μ A	$\ln(I/\mu\text{A})$
0.490	1	
0.568	10	
0.647	100	
0.732	1000	
0.818	5000	

- (i) Complete Table 2 by filling in the missing values. [1 mark]
- (ii) Plot a graph of $\ln I$ versus V on the grid opposite . Draw your best straight line through the points. [4 marks]
- (iii) Write the equation relating $\ln I$ and V . [1 mark]
- (iv) Determine the ideality factor n .

[4 marks]

[Data: $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$]

Total 15 marks



3. (a) The radioactive gas Radon-220, an α -emitter, is easily separated from its powdered solid parent and is known to have a short half-life. Describe how you would accurately measure the half-life of Radon. In your account give details of the apparatus required, the procedure to be followed and the way the collected data will be processed.

[illegible]

[7 marks]

- (b) The half-life of Radon-220 is actually 54 seconds: it decays by the emission of α -particles, one particle being emitted for each atom decaying.

- (i) How many α -particles would be emitted by a 4 mg sample of $^{220}_{86}\text{Rn}$ in 108 seconds?

[5 marks]

- (ii) A laboratory is deemed safe after a Radon-220 leak when the activity has decreased to less than 0.1% of its original value. What is the MINIMUM number of minutes for which the laboratory must remain closed?

[3 marks]

Total 15 marks

SECTION B

Answer ALL questions in this section.

Write your answers in the spaces provided in this answer booklet.

4. (a) State Kirchhoff's two laws for electrical circuits and give the physical principle that each law is based on. [4 marks]
- (b) Distinguish between the 'e.m.f.' and the 'terminal p.d.' of a cell. [2 marks]
- (c) Figure 3 shows a network of resistors and batteries. The internal resistance of each battery is 1Ω as shown.

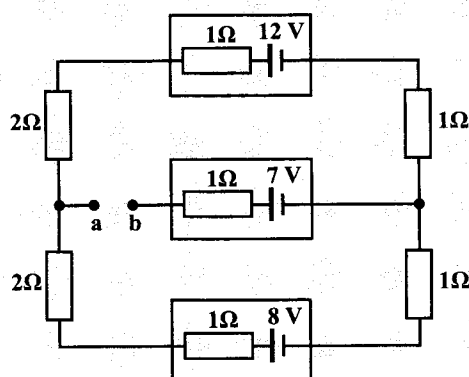


Figure 3

- (i) Calculate the current flowing through the 12 V battery.
- (ii) Points *a* and *b* are connected by a wire of negligible resistance. Calculate the new value for the current that will flow through the 12 V battery. [9 marks]

Total 15 marks

Write your answer to Question 4 here.

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一、二、三、四、五、六、七、八、九、十、十一、十二、十三、十四、十五、十六、十七、十八、十九、二十、二十一、二十二、二十三、二十四、二十五、二十六、二十七、二十八、二十九、三十、三十一、三十二、三十三、三十四、三十五、三十六、三十七、三十八、三十九、四十、四十一、四十二、四十三、四十四、四十五、四十六、四十七、四十八、四十九、五十、五十一、五十二、五十三、五十四、五十五、五十六、五十七、五十八、五十九、六十、六十一、六十二、六十三、六十四、六十五、六十六、六十七、六十八、六十九、七十、七十一、七十二、七十三、七十四、七十五、七十六、七十七、七十八、七十九、八十、八十一、八十二、八十三、八十四、八十五、八十六、八十七、八十八、八十九、九十、九十一、九十二、九十三、九十四、九十五、九十六、九十七、九十八、九十九、一百。

5. Figure 4 shows an ideal non-inverting operational amplifier (op-amp) circuit.

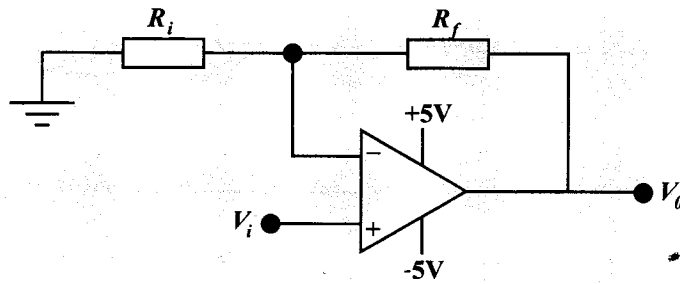


Figure 4

- (a) Show that the closed loop gain, A , of the circuit is given by $A = \frac{R_i + R_f}{R_i}$. Clearly state any assumptions made about the properties of the op-amp. **[5 marks]**
- (b) A circuit like the one in Figure 4 is set up with $R_f = 440 \text{ k}\Omega$ and $R_i = 40 \text{ k}\Omega$. The op-amp is operated from a $\pm 5.0 \text{ V}$ supply.
- What is the value of the output voltage when is $V_i = +250 \text{ mV}$?
 - If the op-amp is NOT to be saturated, what is the MAXIMUM voltage amplitude for the input signal?
 - Sketch a graph to show the expected output when a sinusoidal signal with an amplitude of 0.75 V is applied to the input of this amplifier. **[6 marks]**
- (c) In a certain application, it is desired to combine two signals, v_1 and v_2 , to form a signal v_o according to the relation $v_o = -2 v_1 - 5 v_2$. The minimum input resistance for both signal inputs should be NO less than $10.0 \text{ k}\Omega$. Design a circuit to meet this requirement. **[4 marks]**

Total 15 marks

Write your answer to Question 5 here.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. A small dark speck is located near the top center of the page. The paper appears to be a standard notebook or legal pad style.

6. (a) With the aid of a sketch graph, clearly explain the following terms and how they originate.
- (i) Continuous X-ray spectrum
 - (ii) Characteristic X-ray spectrum
 - (iii) Cut-off wavelength
- [7 marks]**

- (b) The x-ray tube in Figure 5 operates at 50 kV and the current through it is 1.2 mA.

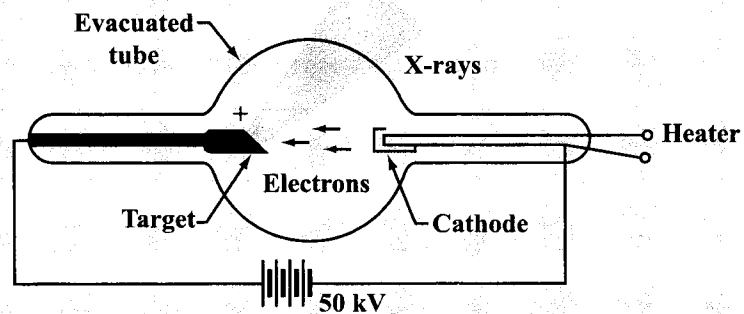


Figure 5

Calculate

- (i) the electrical power input
- (ii) the speed of the electrons when they hit the target
- (iii) the cut-off wavelength of the X-rays emitted.

[8 marks]

Total 15 marks

COAST GUARD VESSEL NO. 10

END OF TEST